

CS3 Rubric

DS 4002 – Spring 2026 - Emily Sun

Individual Assignment

Why am I doing this? This case study gives you the opportunity to apply data science and statistical analysis to a real-world problem at the intersection of politics and financial markets. Specifically, you will investigate whether tariff-related announcements from Donald Trump have a measurable impact on stock market behavior. As you work through this assignment, you will be exposed to how data analysis can be used to evaluate the effects of political events and better understand market dynamics.

What am I going to do? You will use a provided GitHub repository containing data and code to reproduce an analysis. Your task is to run, understand, and critique the full workflow. You will:

- Preprocess and align tariff-related post data with stock market data
- Compare market behavior on days with and without tariff-related announcements, including metrics such as returns and volatility
- Evaluate whether the conclusions are supported by the analysis
- **(Bonus)** Retrieve updated tariff-related posts and assess whether incorporating new data changes the results/conclusions. The original data is in the github but was from FiscalNote.

After completing your analysis, your deliverables will include:

- A GitHub repository containing:
 - Your reproduced and/or updated code
 - Clear documentation of any changes or improvements made
- A written report (PDF) including:
 - Summary of the problem and its importance
 - Critique of the case study (strengths and weaknesses)
 - Interpretation of results and their significance
 - Reflection on what could be improved
- A reference list citing any additional sources used

Tips for success:

- Be bold. This is your chance to pick something you learned and share it.
- Don't overthink it. A clear presentation of fundamentals is more valuable than an unclear presentation of cutting edge techniques.
- Talk to instructors and classmates: this is a creative assignment, and you are encouraged to share ideas for feedback.

How will I know I have Succeeded? You will meet expectations on the case study when you follow the criteria in the rubric below.

Formatting	◦ Writing Portion: Submit this as a pdf document, should be approximately 2-3 pages in length

	- o Coding Portion: Include code under a CODE or SCRIPTS folder in Github. Include any additional or updated code that was used in Github if necessary. - o References: Make sure all references are cited accordingly
Written Portion	• <u>Goal</u>: Discuss your interpretation of the study and thought process by answering the questions below. • a. Problem Summary (1 paragraph) - o What is the study trying to answer? - o Why does it matter in a real-world context? • b. Critique (2 paragraphs) - o One positive critique: What was done well? (data, methods, clarity, etc.) - o One negative critique: What could be improved? What challenges did you encounter while reproducing the analysis? How did you overcome them? • c. Results & Interpretation (2 paragraphs) - o What did the analysis find? Are the results convincing? Why or why not? Do the methods support the conclusions? • d. Reflection & Improvement (2 paragraphs) - o What would you do differently? How could the analysis be improved (data, methods, assumptions)? What would be your next step if continuing this project?
Code	• <u>Goal</u>: Reproduce the code, clearly document the process, and consider potential improvements or extensions. • There should be: - o Reproducible Code: code used to perform preprocessing, exploratory data analysis, and analysis, code runs from start to finish without errors • Documentation - o Clear comments explaining key steps, transformations, and decisions, logical organization of scripts or notebooks • Outputs - o Data visualizations (graphs) and/or tables showing results - o Clearly labeled and interpretable outputs • Bonus: - o Updated data validation: At the time of reproduction, additional tariff-related posts may exist. Attempt to retrieve updated data and evaluate whether incorporating new observations changes the results or conclusions.
References	• At the end of the written portion of the assignment, please include a list of references that were not already given in the materials.

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